CHAPTER 4

THEORY OF THE FIRM UNDER PERFECT COMPETITION

CHAPTER ESSENCE AT A GLANCE:
Master perfectly competitive market dynamics: price-taking behavior, horizontal revenue curves (P = AR = MR), short-run profit maximization conditions, and supply curve derivation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Features of Perfect Competition:** Characterized by very large numbers of buyers and sellers, homogeneous products, free entry and exit, perfect factor mobility, and perfect market information, making individual firms price takers.
- **Horizontal Revenue Curves:** Because the firm takes the market equilibrium price as given, Average Revenue (AR) and Marginal Revenue (MR) are identical, equal to market price P, forming a perfectly elastic horizontal line.
- **Profit Maximization Conditions:** A competitive firm maximizes profit where: (1) P = MC; (2) MC is non-decreasing (rising); and (3) In the short run, P >= AVC (above shut-down point).
- **Supply Curve Derivation:** The short-run supply curve of a competitive firm is the rising portion of its Short-Run Marginal Cost (SMC) curve on or above the minimum Average Variable Cost (AVC) curve.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Competitive Firm Revenue} \iff P = \text{AR} = \text{MR} \rightarrow \text{Horizontal Demand Curve}$$

Price-Taker Perfect Elasticity Condition

$$\text{Profit Maximization} \iff [P = \text{SMC}, \quad \text{SMC is Rising}, \quad P \geq \text{AVC}_{\min}]$$

Three Simultaneous Short-Run Equilibrium Criteria

$$\text{Shut-Down Point} \iff P = \text{AVC}_{\min} \quad \text{vs} \quad \text{Break-Even Point} \iff P = \text{AC}_{\min}$$

Critical Operating Boundaries for Firm Survival

$$e_S = \frac{\%\Delta Q_S}{\%\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Price Elasticity of Supply Responsiveness Measure

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** A firm does not shut down immediately when incurring losses; as long as price covers Average Variable Cost (P >= AVC), it continues operating in the short run to recover part of fixed costs.
- **Alert:** The shut-down point occurs where price equals minimum AVC; the break-even point occurs where price equals minimum AC.
- **Alert:** Market supply curve is obtained by horizontal summation of individual supply curves of all firms in the industry.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State any three characteristic features of a perfectly competitive market and explain why a firm is a price taker. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Three Features:** (1) Very large number of buyers and sellers; (2) Homogeneous standardized products; (3) Free entry and exit of firms in the industry. [1.5 Marks]

• **Why Firm is a Price Taker:** An individual firm produces an infinitesimally small fraction of total market output; any attempt to charge a price above market price results in zero sales, making the firm take the industry equilibrium price as given. [1.5 Marks]

2. Why are Average Revenue (AR) and Marginal Revenue (MR) curves identical and horizontal under Perfect Competition? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Price Taker Mechanism:** Since the firm can sell any quantity of output at the prevailing market price P, price remains constant for all levels of output. [1 Mark]

• **Mathematical Identity:** AR = Total Revenue divided by Output = (P * Q) divided by Q = P. Similarly, each additional unit sold adds exactly P to Total Revenue, so MR = P. [1 Mark]

• **Geometric Shape:** Because AR = MR = P at all output levels, the demand curve faced by the firm is a horizontal straight line parallel to the X-axis with infinite price elasticity. [1 Mark]

3. Explain the three conditions for profit maximization of a competitive firm in the short run. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Condition 1 (Price equals Marginal Cost):** P = SMC. If P > SMC, producing more units adds more to revenue than cost, increasing profit. If P < SMC, expanding output reduces profit. [1.5 Marks]

• **Condition 2 (SMC must be rising):** SMC must be upward sloping at equilibrium. If SMC is falling at P = SMC, further expansion generates positive net gains, making it a point of minimum profit rather than maximum. [1 Mark]

• **Condition 3 (Price covers AVC):** In the short run, P must be greater than or equal to minimum AVC (P >= AVC). If P falls below minimum AVC, the firm minimizes losses by producing zero units. [1.5 Marks]

4. Derive the short-run supply curve of a perfectly competitive firm. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Case 1 (Price below minimum AVC):** When market price P < minimum AVC, the firm cannot cover variable costs, producing zero output. Supply is zero. [1 Mark]

• **Case 2 (Price at or above minimum AVC):** When P >= minimum AVC, the firm equates P = SMC to maximize profit, with output determined by the rising SMC curve. [1 Mark]

• **Conclusion:** The short-run supply curve is the upward sloping segment of the Short-Run Marginal Cost (SMC) curve on and above the minimum Average Variable Cost (AVC) curve. [1 Mark]

5. Distinguish between the Shut-Down Point and the Break-Even Point of a firm. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Shut-Down Point:** The situation where market price equals minimum Average Variable Cost (P = minimum AVC). Below this price, the firm shuts down production in the short run. [1 Mark]

• **Break-Even Point:** The situation where market price equals minimum Average Cost (P = minimum AC). At this point, the firm earns normal profit with zero supernormal economic profits. [1 Mark]

6. When the market price of a commodity increases from Rs 4 to Rs 5 per unit, a firm increases its supply from 100 units to 125 units. Calculate price elasticity of supply. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Given Data:** Initial Price P = 4, New Price P1 = 5; Delta P = 5 - 4 = +1. Initial Supply Q = 100, New Supply Q1 = 125; Delta Q = 125 - 100 = +25. **[1 Mark]**

• **Formula:** eS = (Delta Q divided by Delta P) * (P divided by Q). **[1 Mark]**

• **Calculation:** eS = (25 divided by 1) * (4 divided by 100) = 25 * 0.04 = 1.0. Supply elasticity is unitary elastic (eS = 1). **[1 Mark]**